

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 2
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Section: B

1. TITLE

HTML5 Graphics and CSS3 Web Styling

2. AIM

To understand how CSS3 enhances the appearance of web pages and to explore HTML5 graphics using the Canvas element by designing a visually styled webpage.

3. OBJECTIVE

The objectives of this experiment are:

- To draw basic shapes and text on a webpage using the HTML5 Canvas element.
- To understand CSS3 syntax and different types of selectors.
- To apply colors, fonts, borders, margins and padding to page elements.
- To use CSS3 backgrounds, gradients, box-shadow, text-shadow and rounded corners.
- To implement hover effects on buttons and navigation links.

4. THEORY

The HTML5 `<canvas>` element provides a drawable region on a webpage that can be controlled using JavaScript. Its 2D context object exposes methods such as `strokeRect()`, `arc()`, `moveTo()/lineTo()` and `fillText()` to draw rectangles, circles, lines and text respectively.

CSS3 (Cascading Style Sheets, Level 3) is used to control the presentation of HTML elements. Selectors (element, class, id) target specific elements, while properties such as `background`, `border-radius`, `box-shadow`, `text-shadow` and `linear-gradient()` control their appearance. The `:hover` pseudo-class applies styles only while the mouse pointer is over an element, and combined with the `transition` property it produces smooth animated effects such as a card lifting or a button changing color.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare the HTML5 document type.
2. Add a `<canvas>` element inside the body and give it an id.
3. In a script block, get the 2D drawing context of the canvas using `getContext("2d")`.
4. Draw a rectangle using `strokeRect()`.
5. Draw a circle using `arc()` with `beginPath()` and `stroke()`.
6. Draw a straight line using `moveTo()` and `lineTo()`.
7. Display the student's name using `fillText()`.
8. Build the page layout with a header, navigation bar, three content cards and a footer.
9. Apply CSS3 background colors and a gradient background to the header.
10. Style the cards with borders, border-radius, padding, margin and box-shadow.
11. Add a text-shadow to the header title and apply different font families.
12. Add hover effects on the navigation links and on the buttons using `:hover` and `transition`.
13. Save the file, open it in a browser and verify the drawing and styling.

6. PROGRAM

index.html

```
<!DOCTYPE html>
```

```

<html lang="en">
<head>
<meta charset="UTF-8">
<title>My Styled Webpage</title>
<style>
  * { box-sizing: border-box; }
  body {
    margin: 0;
    font-family: 'Segoe UI', Arial, sans-serif;
    background: #f2f4f8;
    color: #222;
  }

  header {
    background: linear-gradient(135deg, #1a3c6e, #4a90e2);
    color: #fff;
    padding: 25px 30px;
    text-shadow: 2px 2px 4px rgba(0,0,0,0.4);
  }
  header h1 { margin: 0; font-family: 'Georgia', serif; font-size: 32px; }

  nav {
    background: #14305a;
    padding: 12px 30px;
  }
  nav a {
    color: #fff;
    margin-right: 22px;
    text-decoration: none;
    font-weight: bold;
    font-family: Arial, sans-serif;
    padding: 6px 10px;
    border-radius: 4px;
    transition: background 0.3s ease, color 0.3s ease;
  }
  nav a:hover {
    background: #ffffff;
    color: #14305a;
  }

  main { padding: 30px; }

  .canvas-box {
    background: #fff;
    border: 1px solid #ccc;
    border-radius: 10px;
    padding: 15px;
    box-shadow: 0 4px 10px rgba(0,0,0,0.15);
    display: inline-block;
    margin-bottom: 30px;
  }

  .cards {
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
  }
  .card {
    flex: 1;
    min-width: 220px;
    background: #ffffff;
    border: 1px solid #ddd;
    border-radius: 12px;
    padding: 20px;
    box-shadow: 0 4px 8px rgba(0,0,0,0.12);
    text-shadow: 1px 1px 1px rgba(0,0,0,0.05);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
  }
  .card:hover {
    transform: translateY(-6px);
    box-shadow: 0 10px 20px rgba(0,0,0,0.2);
  }
  .card h3 { margin-top: 0; color: #1a3c6e; }
  .card.blue { background: linear-gradient(160deg, #e8f0fe, #ffffff); }
  .card.green { background: linear-gradient(160deg, #e6f9ed, #ffffff); }
  .card.orange { background: linear-gradient(160deg, #fff2e0, #ffffff); }

  button.styled-btn {
    background: #1a3c6e;
    color: #fff;
    border: none;
    padding: 10px 22px;
    border-radius: 6px;
    font-size: 15px;
    cursor: pointer;
    box-shadow: 0 3px 6px rgba(0,0,0,0.2);
    transition: background 0.3s ease, transform 0.2s ease;
  }

```

```

}
button.styled-btn:hover {
  background: #4a90e2;
  transform: scale(1.05);
}

footer {
  background: #14305a;
  color: #fff;
  text-align: center;
  padding: 16px;
  margin-top: 30px;
}
</style>
</head>
<body>

<header>
  <h1>My Styled Webpage</h1>
</header>

<nav>
  <a href="#canvas">Canvas</a>
  <a href="#cards">Cards</a>
  <a href="#contact">Contact</a>
</nav>

<main>
  <section id="canvas">
    <h2>HTML5 Canvas Drawing</h2>
    <div class="canvas-box">
      <canvas id="myCanvas" width="400" height="220"></canvas>
    </div>
    <script>
      const c = document.getElementById('myCanvas');
      const ctx = c.getContext('2d');

      // Rectangle
      ctx.strokeStyle = '#1a3c6e';
      ctx.lineWidth = 3;
      ctx.strokeRect(20, 20, 120, 70);

      // Circle
      ctx.beginPath();
      ctx.arc(230, 55, 40, 0, Math.PI * 2);
      ctx.strokeStyle = '#c0392b';
      ctx.stroke();

      // Line
      ctx.beginPath();
      ctx.moveTo(20, 140);
      ctx.lineTo(300, 140);
      ctx.strokeStyle = '#27ae60';
      ctx.lineWidth = 4;
      ctx.stroke();

      // Name text
      ctx.font = '24px Georgia';
      ctx.fillStyle = '#222';
      ctx.fillText('K. Sirisha', 20, 190);
    </script>
  </section>

  <section id="cards">
    <h2>Content Cards</h2>
    <div class="cards">
      <div class="card blue">
        <h3>Card One</h3>
        <p>Styled with gradients, rounded corners and a soft box shadow.</p>
        <button class="styled-btn">Learn More</button>
      </div>
      <div class="card green">
        <h3>Card Two</h3>
        <p>Hover over this card to see the lift and shadow transition effect.</p>
        <button class="styled-btn">Learn More</button>
      </div>
      <div class="card orange">
        <h3>Card Three</h3>
        <p>Demonstrates borders, padding, margin and text-shadow together.</p>
        <button class="styled-btn">Learn More</button>
      </div>
    </div>
  </section>
</main>

<footer id="contact">
  &copy; 2026 My Styled Webpage – CSS3 & HTML5 Canvas Demo

```

```
</footer>
```

```
</body>
```

```
</html>
```

7. CODE EXPLANATION

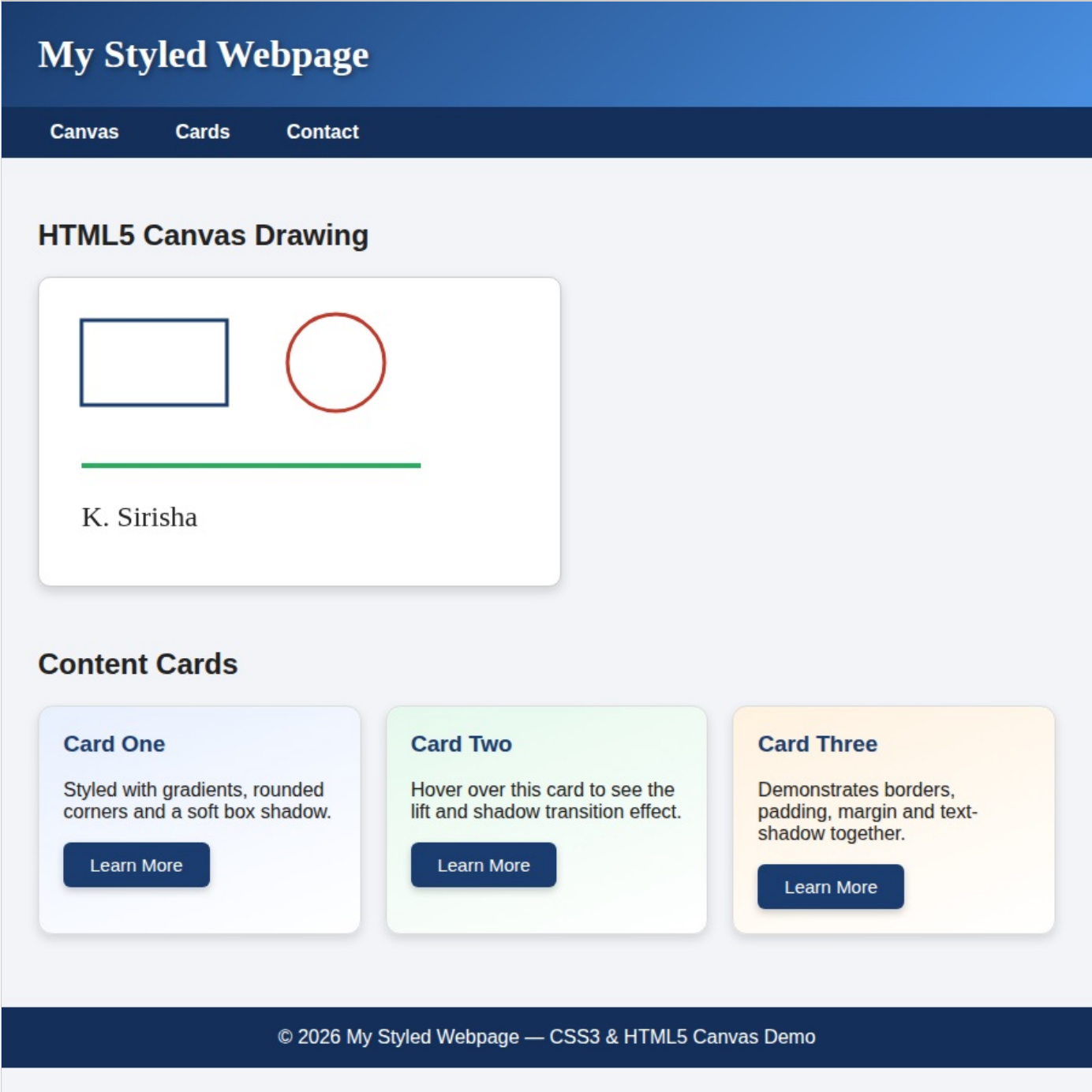
Code	Explanation
<code><canvas id="myCanvas"></code>	Reserves a drawable area on the page.
<code>getContext("2d")</code>	Returns the 2D drawing context used to draw shapes.
<code>strokeRect(x,y,w,h)</code>	Draws the outline of a rectangle at the given position and size.
<code>arc(x,y,r,0,2*PI) + stroke()</code>	Draws a circle of radius r centred at (x, y).
<code>moveTo() / lineTo()</code>	Draws a straight line between two points.
<code>fillText()</code>	Renders text (the student's name) onto the canvas.
<code>linear-gradient()</code>	Creates a smooth gradient background for the header.
<code>box-shadow</code>	Adds a soft drop shadow around the cards and canvas box.
<code>text-shadow</code>	Adds a shadow behind the header text and card text.
<code>border-radius</code>	Rounds the corners of the cards, canvas box and buttons.
<code>:hover + transition</code>	Animates nav links and buttons smoothly when the mouse hovers over them.

8. TEST CASES AND EXECUTION DATA

Test Case	Action / Input	Expected Result	Actual Result	Status
TC-01	Load the page in browser	Header, navigation bar, canvas and cards display correctly	Displayed correctly	Pass
TC-02	Inspect the canvas area	Rectangle, circle, line and name text are visible	All four elements rendered	Pass
TC-03	Hover over a navigation link	Link background changes with a smooth transition	Background changed smoothly	Pass
TC-04	Hover over a content card	Card lifts up with an enhanced shadow	Lift and shadow effect worked	Pass
TC-05	Hover over "Learn More" button	Button color changes and scales slightly	Color changed and button scaled	Pass
TC-06	Resize the browser window	Cards wrap responsively without overlapping	Cards wrapped correctly	Pass

9. OUTPUT

Output: Canvas Drawing and Styled Content Cards



Rendered in browser — HTML5 Canvas showing a rectangle, circle, line and name text; gradient header with text-shadow; navigation bar; three content cards with rounded corners, box-shadow; styled buttons and footer.

10. RESULT

Thus, a webpage titled "My Styled Webpage" was successfully developed, demonstrating HTML5 Canvas drawing (rectangle, circle, line and text) along with CSS3 styling features such as gradients, shadows, rounded corners and hover transition effects. The output was verified against the expected results for all test cases.